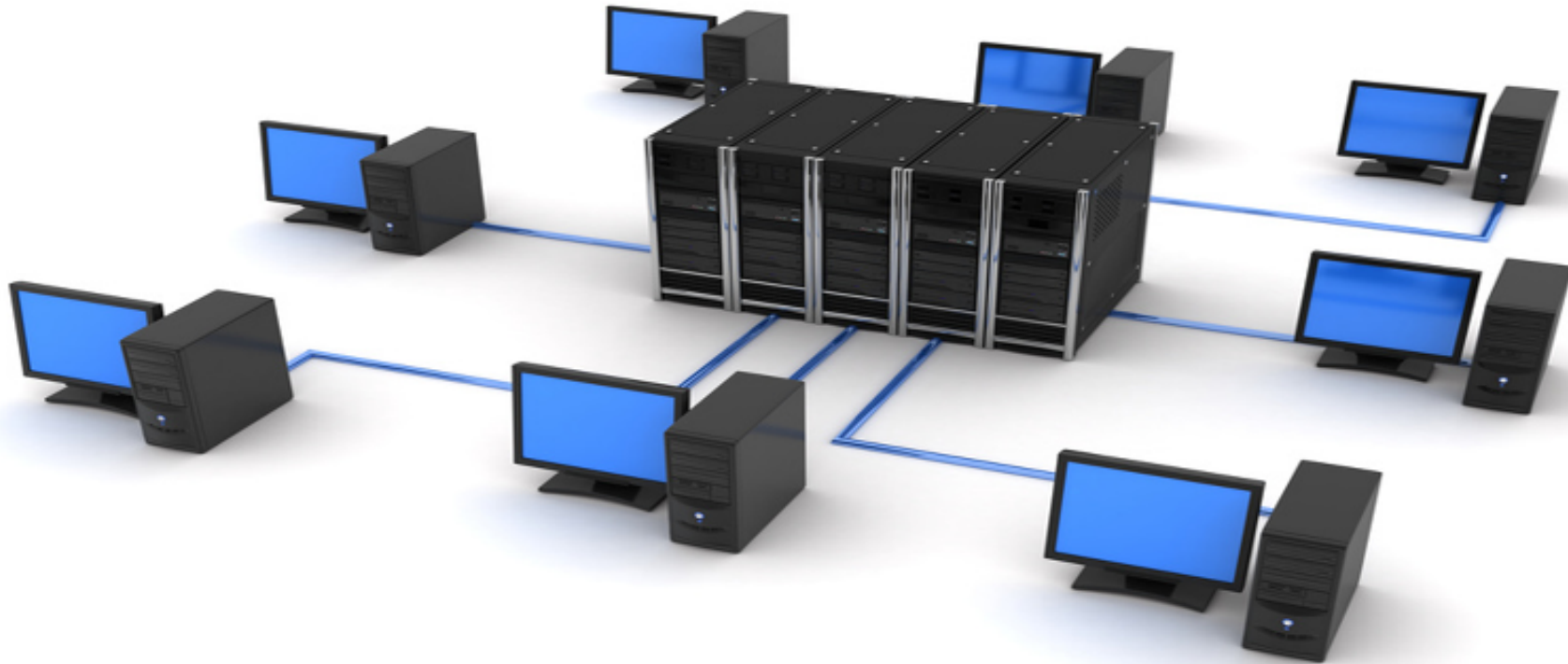


Chapter 8: Database Management Systems (DBMS)



Topic Objectives

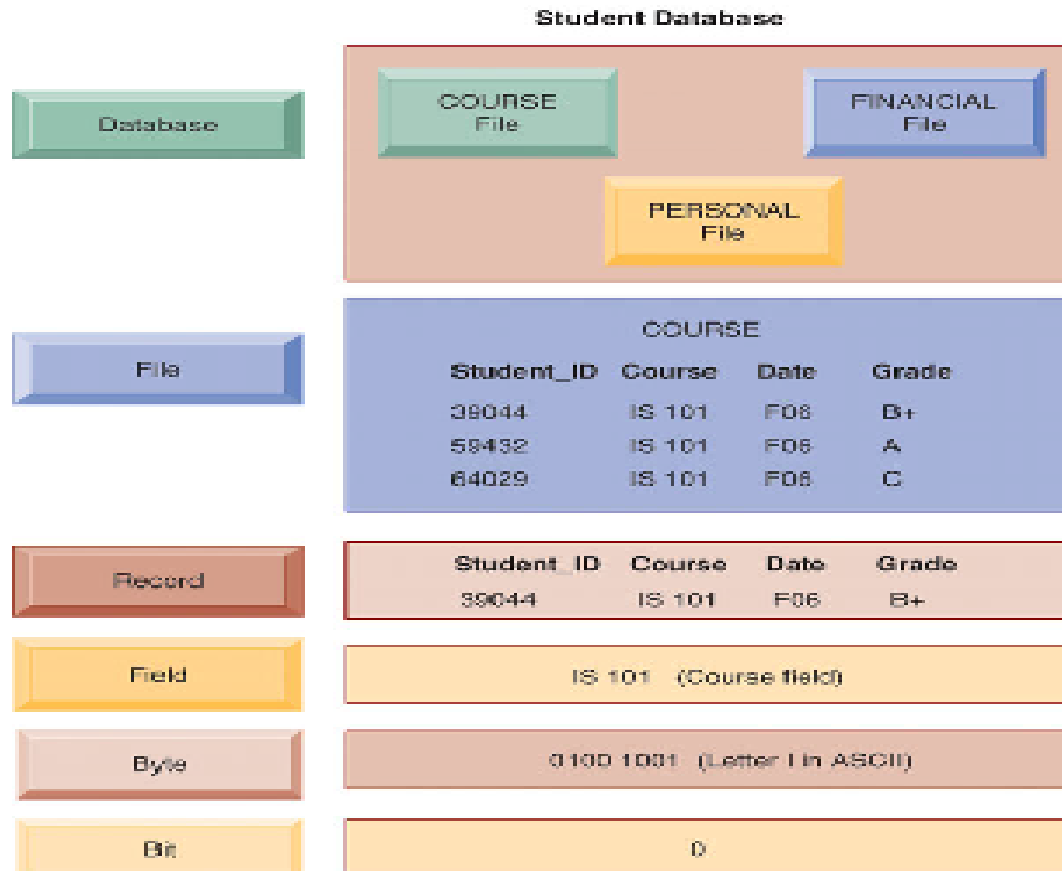
- Database systems vs. file-based systems
- Meaning of the terms.
- DB languages
- The three-level database architecture & its purpose.
- Meaning of logical and physical data independence.
- Major functions & components of the DBMS environment.
- Roles in the DB environments.
- Advantages and disadvantages of DBMSs.
- Lifecycle of DBS development

Database Systems vs. File-based Systems

- Effective information system provides users with accurate, timely, and relevant information. Accurate information is free of errors.
- A computer system organizes data in a hierarchy that starts with bits and bytes and progresses to fields, records, files, and databases
- A bit represents the smallest unit of data a computer can handle. A group of bits, called a byte, represents a single character, which can be a letter, a number, or another symbol.
- A grouping of characters into a word, a group of words, or a complete number (such as a person's name or age) is called a field. A group of related fields, such as the student's name, the course taken, the date, and the grade, comprises a record; a group of records of the same type is called a file.

Database Systems vs. File-based Systems

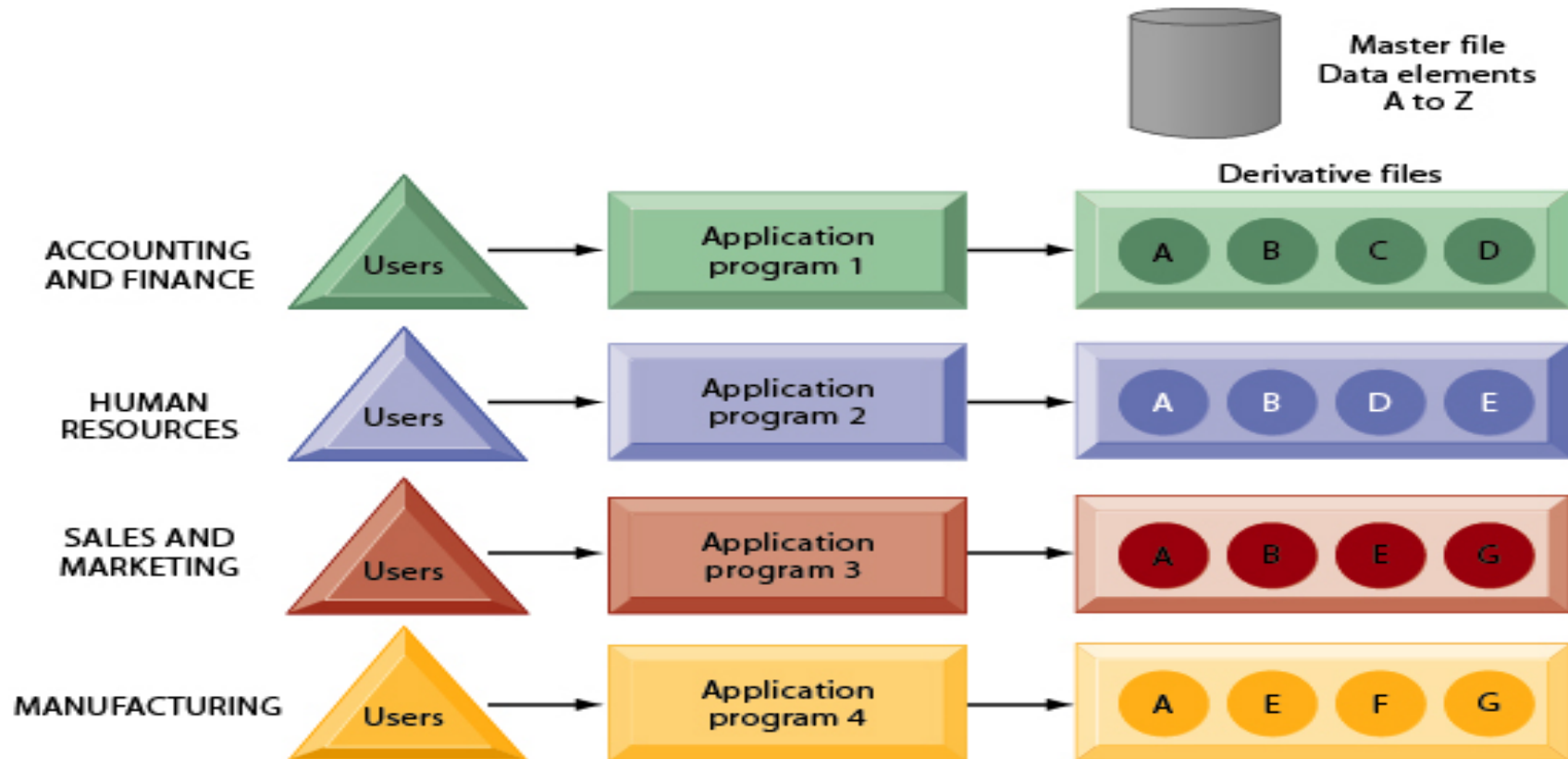
The Data Hierarchy



Traditional File Processing

- The use of a traditional approach to file processing encourages each functional area in a corporation to develop specialized applications.
- Each application requires a unique data file that is likely to be a subset of the master file.
- Each application, of course, required its own files and its own computer program to operate.
- For example, the human resources functional area might have a personnel master file, a payroll file, a medical insurance file, a pension file, a mailing list file, and so forth until tens, perhaps hundreds, of files and programs existed.

Traditional File Processing



Limitations of File-based Approach

- **Separation and isolation of data-** Each program maintains its own set of data. Users of one program may be unaware of potentially useful data held by other programs.
- **Data Redundancy and Inconsistency-** Presence of duplicate data in multiple data files so that the same data are stored in more than place or location. Leads to wastage of storage resources and also leads to data inconsistency, where the same attribute may have different values.
- **Program-Data Dependence -** Coupling of data stored in files and the specific programs required to update and maintain those files such that changes in programs require changes to the data. Every traditional computer program has to describe the location and nature of the data with which it works

Limitations of File-based Approach

- **Lack of Flexibility** - A traditional file system can deliver routine scheduled reports after extensive programming efforts, but it cannot deliver ad hoc reports or respond to unanticipated information requirements in a timely fashion.
- **Incompatible File Formats** - Programs are written in different languages, and so cannot easily access each others files.
- **Poor Security**- Because there is little control or management of data, access to and dissemination of information may be out of control. Management may have no way of knowing who is accessing or even making changes to the organization's data
- **Lack of Data Sharing and Availability** - Because pieces of information in different files and different parts of the organization cannot be related to one another, it is virtually impossible for information to be shared or accessed in a timely manner

The Database Approach

- Database technology cuts through many of the problems of traditional file organization.
- A more rigorous definition of a database is a collection of data organized to serve many applications efficiently by centralizing the data and controlling redundant data. Rather than storing data in separate files for each application, data are stored so as to appear to users as being stored in only one location.
- A single database services multiple applications. For example, instead of a corporation storing employee data in separate information systems and separate files for personnel, payroll, and benefits, the corporation could create a single common human resources database.

Database Various Definitions

- Shared collection of logically related data (and a description of this data), designed to meet the information needs of an organization.
- System catalog (metadata) provides description of data to enable program–data independence.
- Organized, shared, self-describing collection of logically related data

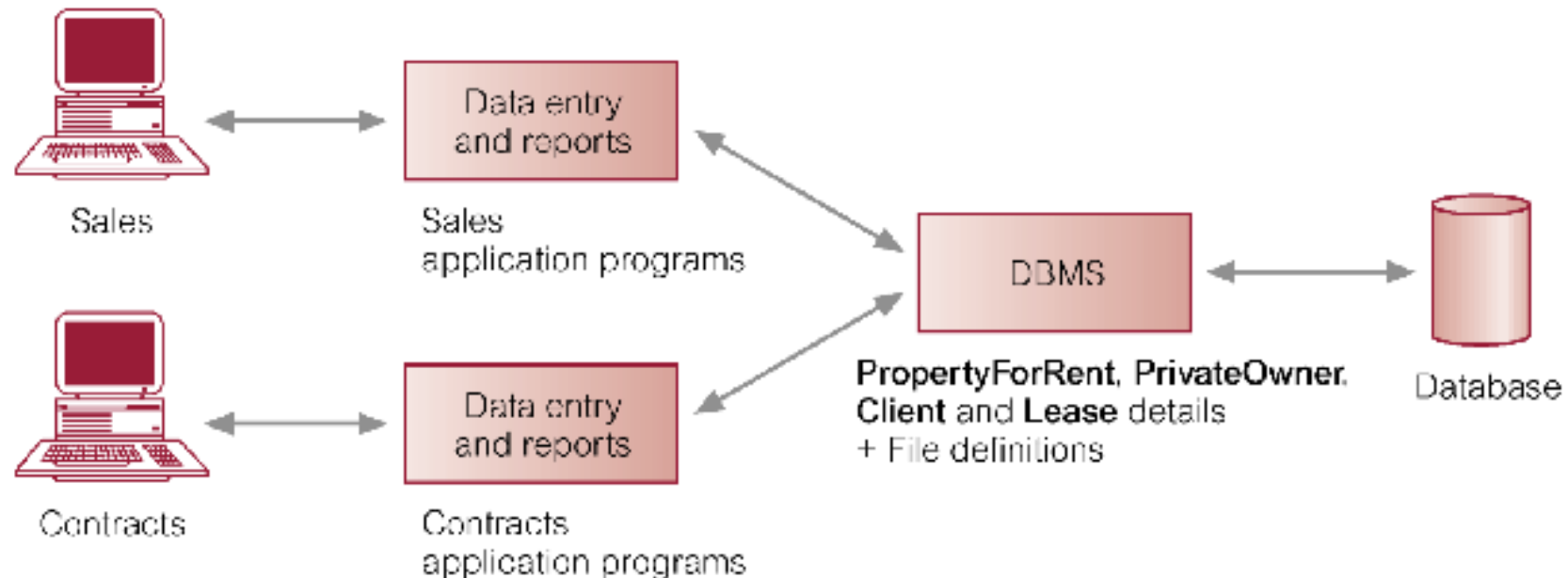
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Database Management System (DBMS)

- A database management system (DBMS) is software that permits an organization to centralize data, manage them efficiently, and provide access to the stored data by application programs.
- The DBMS acts as an interface between application programs and the physical data files.
- When the application program calls for a data item, such as gross pay, the DBMS finds this item in the database and presents it to the application program

Database Management System (DBMS)



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

PrivateOwner (ownerNo, fName, lName, address, telNo)

Client (clientNo, fName, lName, address, telNo, preType, maxRent)

Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentFinish)

Database Approach & Languages

- **Data definition language (DDL)** - Permits specification of data types, structures and any data constraints. All specifications are stored in the database. Provides basic data manipulation operations on data held in the database
- **Data manipulation language (DML)**- Includes tools for accessing and manipulating information in databases by allowing users to add, change, delete, and retrieve the data in the database. This language contains commands that permit end users and programming specialists to extract data from the database to satisfy information requests. The most prominent data manipulation language today is Structured Query Language, or SQL
- **Data Control language (DCL)** - General administration & security facility (same syntax as query language) used mainly by the DBA.

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Database Approach and Control

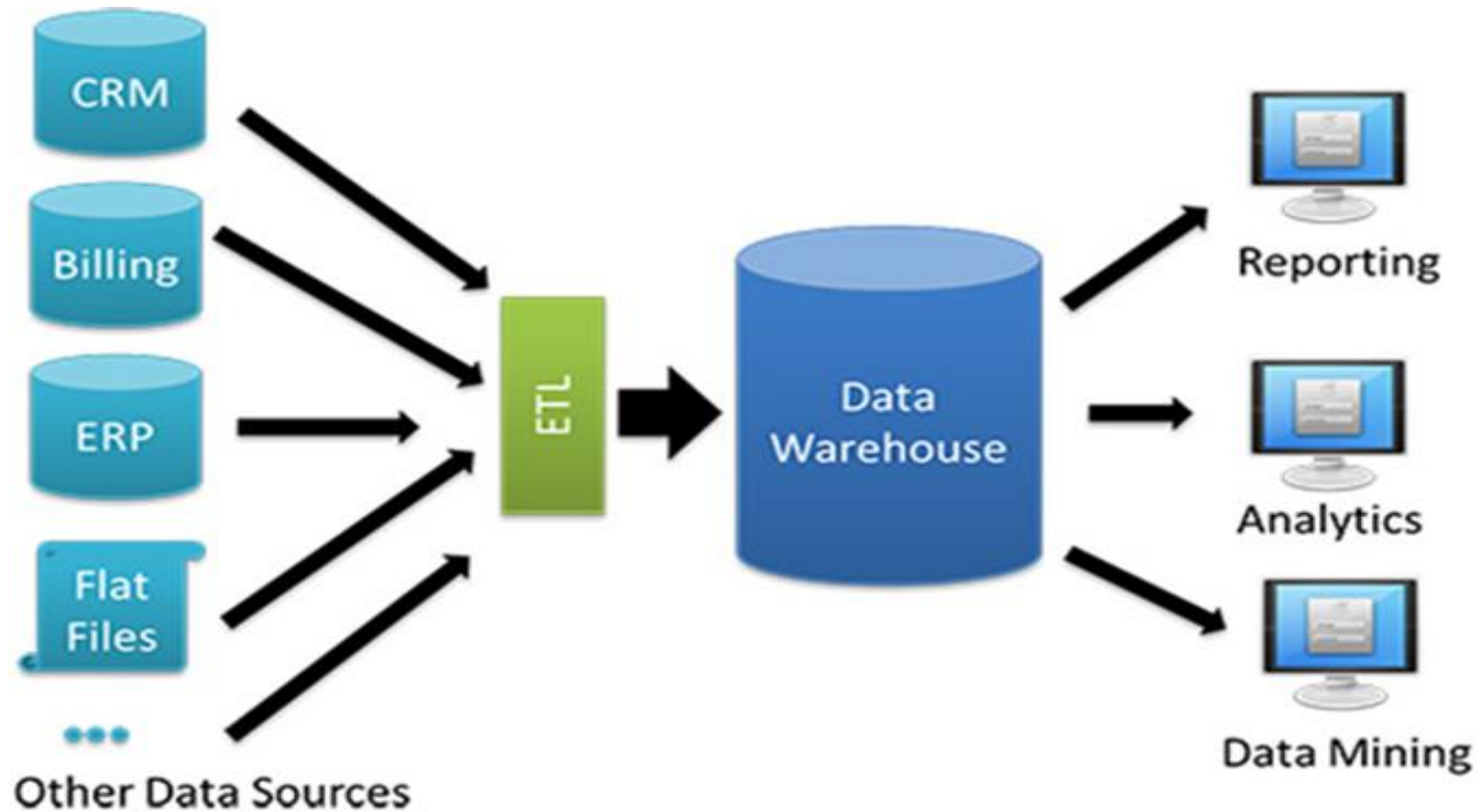
➤ Controlled access to database may include:

- ☐ A security system.
- ☐ An integrity system.
- ☐ A concurrency control system.
- ☐ A recovery control system.
- ☐ A user-accessible catalog.

Data Warehouse

- A data warehouse is a database that stores current and historical data of potential interest to decision makers throughout the company.
- The data originate in many core operational transaction systems, such as systems for sales, customer accounts, and manufacturing, and may include data from Web site transactions.
- The data warehouse consolidates and standardizes information from different operational databases so that the information can be used across the enterprise for management analysis and decision making.

Data Warehouse

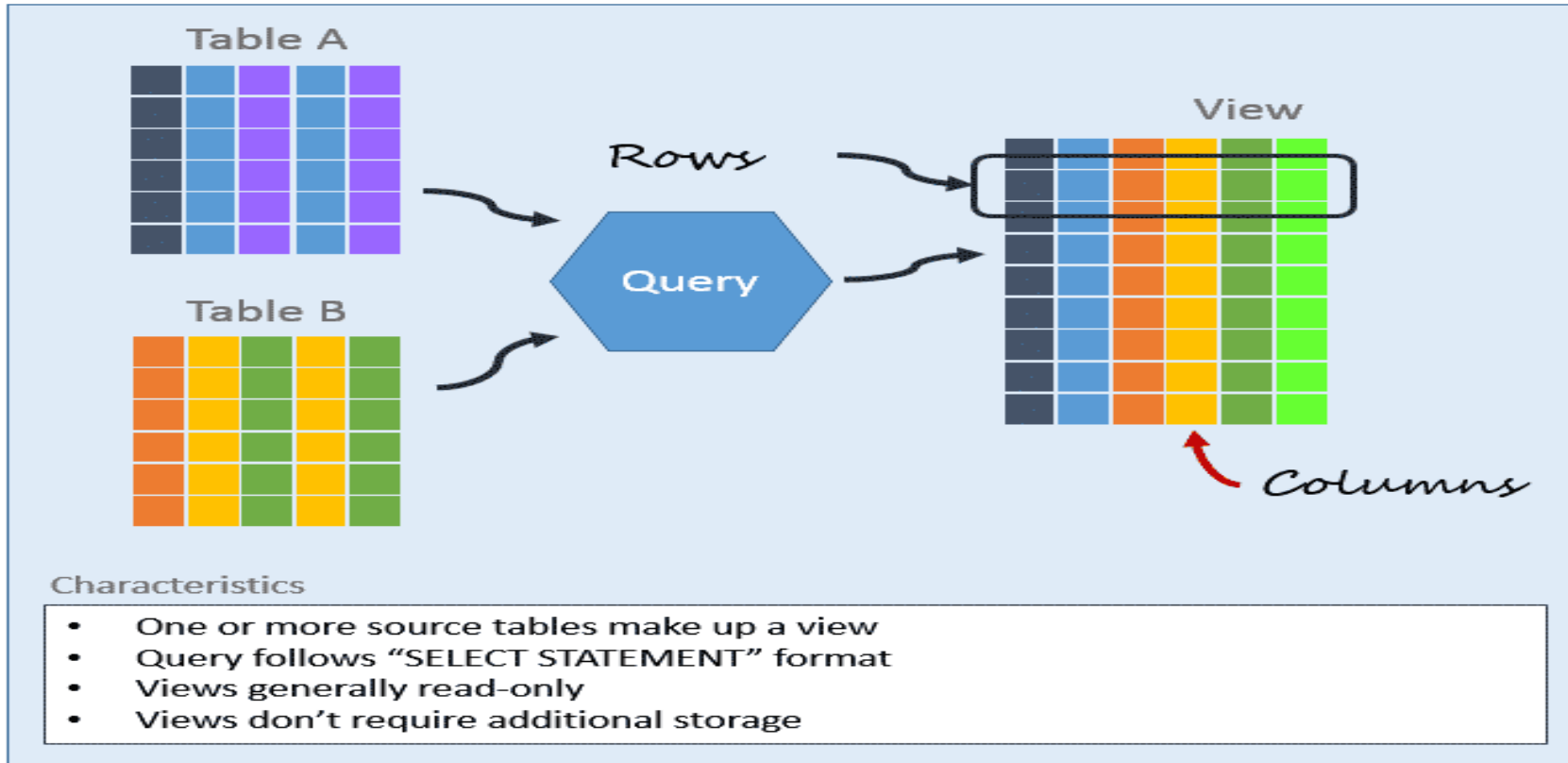


Database View

- A view mechanism provides users with only the data they want or need to use
- Allows each user to have his or her own view of the database. A view is essentially some subset of the database
- A database view displays one or more database records on the same page. A view can display some or all of the database fields
- Views have filters to determine which records they show. Views can be sorted to control the record order and grouped to display records in related sets. Views have other options such as totals and subtotals.

Database View

Anatomy of a View



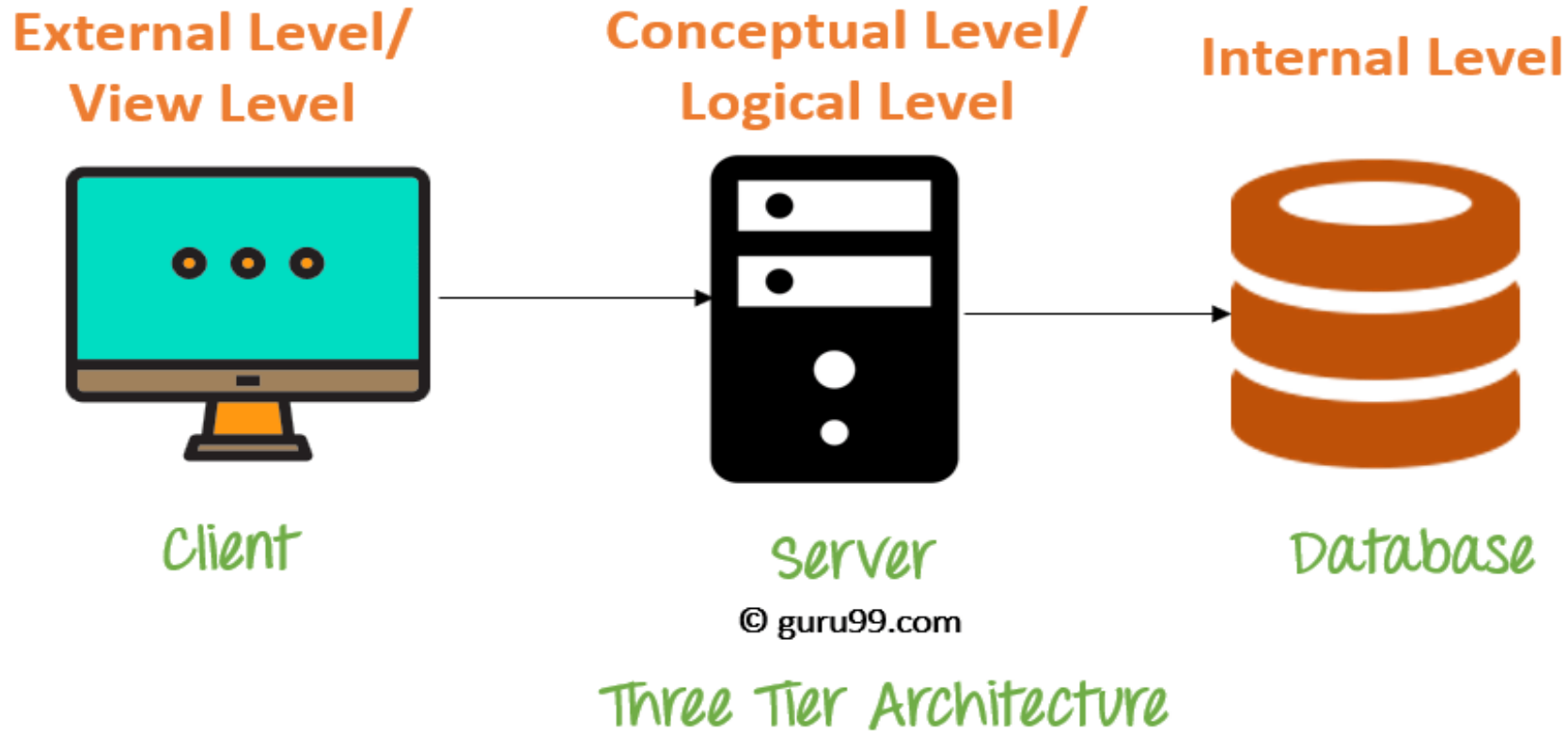
Benefits of Database Views

- Provide additional level of table security by restricting access to a predetermined set of rows or columns of a table
- Hide Data complexity: For example, a single view might be defined with a join, which is a collection of related columns or rows in multiple tables. However, the view hides the fact that this information actually originates from several tables
- Provide a mechanism to customize the appearance of the database;
- Present a consistent, unchanging picture of the structure of the database, even if the underlying database is changed.
- Isolate applications from changes in definitions of base tables.

Three Level Architecture of DBMS

- Database systems comprise of complex data structures. Thus, to make the system efficient for retrieval of data and reduce the complexity of the users, developers use the method of Data Abstraction.
- There are mainly three levels of data abstraction:
 - ❑ Internal Level / Schema
 - ❑ Conceptual or Logical Level /Schema
 - ❑ External or View level /Schema

ANSI – SPARC Three Tier Architecture of DBMS



Internal Level / Schema

- This is the lowest level in the three level architecture. It is also known as the internal level.
- The physical level describes how data is actually stored in the database.
- In the lowest level, this data is stored in the external hard drives in the form of bits and at a little high level, it can be said that the data is stored in files and folders.
- The physical level also discusses compression and encryption techniques

Conceptual or Logical Level

- The conceptual level is at a higher level than the physical level. It is also known as the logical level.
- The conceptual schema describes the Database structure of the whole database for the community of users.
- This schema hides information about the physical storage structures and focuses on describing data types, entities, relationships, etc.
- The conceptual level does not care for how the data in the database is actually stored.

External or View Level

- The external level only shows the relevant database content to the users in the form of views and hides the rest of the data.
- So different users can see the database as a different view as per their individual requirements.
- An external view is just the content of the database as it is seen by some specific particular user. For example, a user from the sales department will see only sales related data

Objectives of Three-Level Architecture

- DBA should be able to change database storage structures without affecting the users' views.
- Internal structure of database should be unaffected by changes to physical aspects of storage.
- DBA should be able to change conceptual structure of database without affecting all users

Data Dependence of the DBMS

- Data Independence is defined as a property of DBMS that helps you to change the Database schema at one level of a database system without requiring to change the schema at the next higher level.
- Data independence helps you to keep data separated from all programs that make use of it.
- You can use this stored data for computing and presentation. In many systems, data independence is an essential function for components of the system.
- DBMS there are two types of data independence
 - ❑ Physical Data Independence
 - ❑ Logical Data Independence.

Physical Data Dependence

- Physical data independence helps you to separate conceptual levels from the internal/physical levels.
- With Physical independence, you can easily change the physical storage structures or devices with an effect on the conceptual schema
- Examples of changes under Physical data dependence include
 - ❑ Using a new storage device like Hard Drive or Magnetic Tapes
 - ❑ Modifying the file organization technique in the Database
 - ❑ Switching to different data structures.
 - ❑ Changing the access method.
 - ❑ Modifying indexes.
 - ❑ Changes to compression techniques or hashing algorithms.
 - ❑ Change of Location of Database from say C drive to D Drive

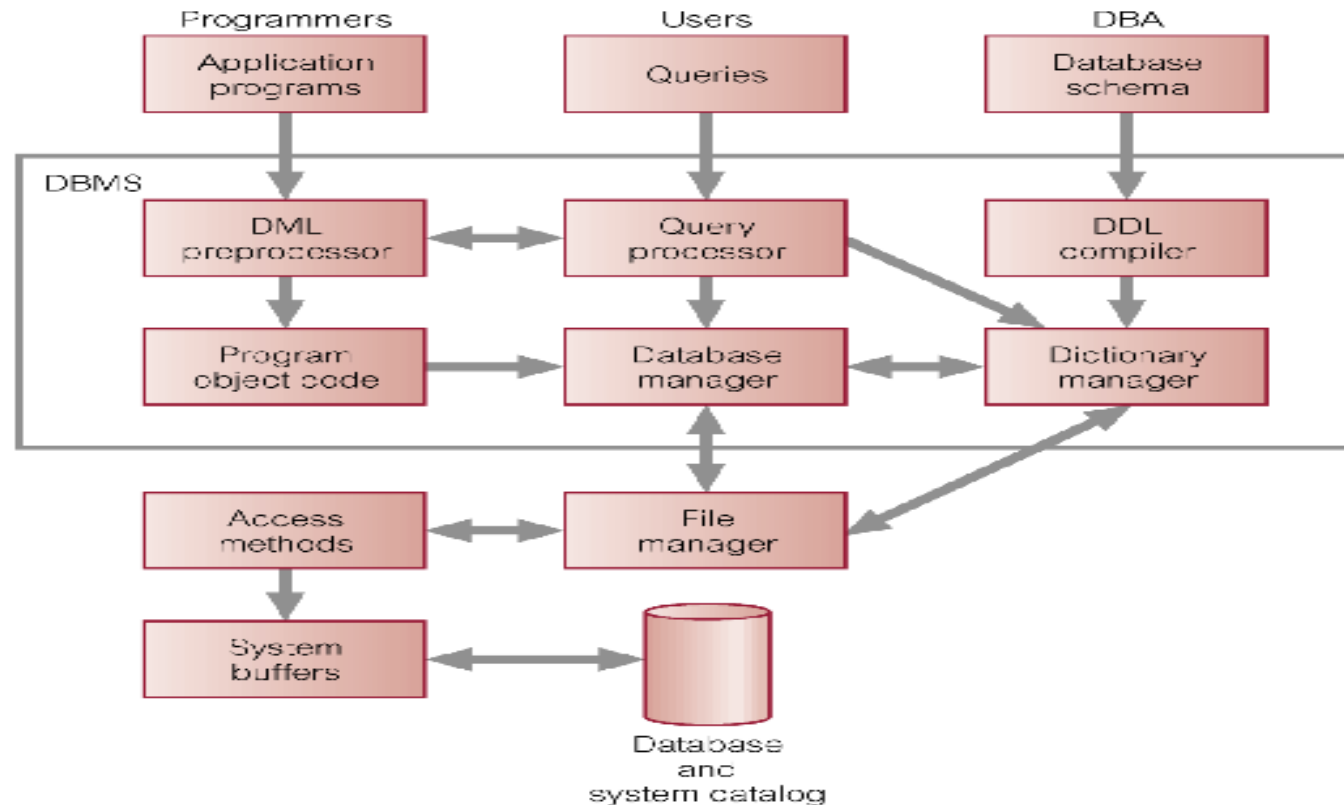
Logical Data Dependence

- Logical data independence is a kind of mechanism, which liberalizes itself from actual data stored on the disk.
- If we do some changes on table format, it should not change the data residing on the disk.
- Examples of changes under logical data dependence include
 - ❑ Add/Modify/Delete a new attribute, entity or relationship is possible without a rewrite of existing application programs
 - ❑ Merging two records into one
 - ❑ Breaking an existing record into two or more records

Functions of a DBMS

- Data Storage, Retrieval, and Update.
- A User-Accessible Catalog.
- Transaction Support.
- Concurrency Control Services.
- Recovery Services.
- Authorization Services.
- Support for Data Communication.
- Integrity Services.
- Services to Promote Data Independence.
- Utility Services.

Components of a DBMS



Roles in the Database Environment

- Data Administrator(DA)
- Database Administrator (DBA)
- Database Designers (Logical and Physical)
- Application Programmers
- End Users (naive and sophisticated)

Advantages of DBMS

- Control of data redundancy
- Data consistency
- More information from the same amount of data.
- Sharing of data
- Improved data integrity
- Improved security
- Enforcement of standards
- Economy of scale

Advantages of DBMS(Continued)

- Balanced conflicting requirements
- Improved data accessibility and responsiveness
- Increased productivity
- Improved maintenance through data independence
- Increased concurrency
- Improved backup and recovery services

Disadvantages of DBMS

- Complexity
- Size
- Cost of DBMS
- Additional hardware costs
- Cost of conversion
- Performance
- Higher impact of a failure

When a DBMS may be unnecessary

- If the database and applications are simple, well defined, and not expected to change
- If there are stringent real-time requirements that may not be met because of DBMS overhead.
- If access to data by multiple users is not required.

Database Development Life Cycle

- DB planning
- System definition
- Requirement collection & analysis
- DB design
- DBMS selection
- Application design
- Prototyping
- Implementation
- Data conversion & loading
- Testing
- Operational maintenance

Database Development Life Cycle

